

A New Method for Chebyshev Polynomial Interpolation Based on Cosine Transforms

Bing-Zhao Li^{1,2} · Yan-Li Zhang^{1,2} · Xian Wang^{1,2} ·
Qi-Yuan Cheng¹

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Abstract Interpolation plays an important role in the areas of signal processing and applied mathematics. Among the various interpolation methods, those related to Chebyshev polynomial interpolation have received much interest recently. In this paper, we propose a new interpolation method using a type I discrete cosine transform (type I DCT) and the nonuniform roots of the second type of Chebyshev polynomials. In this method, the interpolation coefficients are derived using the type I DCT of the Chebyshev nonuniform sampling points. Simulations show the correctness of the proposed method, and a comparison of the proposed method with existing methods is also discussed in detail.

Keywords Chebyshev polynomial · Chebyshev nonuniform sampling · Polynomial interpolation · Coefficients · Discrete cosine transform · Error analysis

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✉ Bing-Zhao Li
li_bingzhao@bit.edu.cn

Yan-Li Zhang
783753093@qq.com

Xian Wang
newlifer_fish@yeah.net

Qi-Yuan Cheng
chqiyuan@bit.edu.cn

¹ School of Mathematics and Statistics, Beijing Institute of Technology, Beijing, China

² Beijing Key Laboratory of Fractional Signals and Systems, Beijing, China

1 Introduction

Interpolation plays an important role in modern approximation science. In the mathematical field of numerical analysis, it is a method for constructing new data points within the range of a discrete set of known data points. In engineering and science, it is often necessary to interpolate (i.e. estimate) the value of a function for an intermediate value of an independent variable [15, 24, 28]. Interpolation theory has been developed since the seventeenth century when calculus was proposed. The most important early contributions were the equidistant nodes interpolation formula and average squared difference formula [4, 5, 16]. Nearly half a century since the use of computers has become widespread and as a result of practical needs in signal processing, ship-building, aviation, and processing machinery, interpolation methods have been further developed both in theory and in practice [1, 11, 12, 17, 27, 33].

With the rapid development of modern technologies, many types of interpolation methods have been proposed, including piecewise constant, linear, polynomial, and spline interpolation methods [3, 6, 34]. Among them, interpolation based on Chebyshev polynomials has received much interest and has been shown to be one of the important methods in the literature [2, 7, 14, 18, 23, 29, 30, 32], because this kind of interpolation polynomial minimizes the problem of Runge's phenomenon [8–10, 13, 19].

Function approximation based on Chebyshev polynomial interpolation and the discrete cosine transform (DCT) was proposed in [2, 18, 29, 31, 32]. In these methods, the nonuniform sampling points according to Chebyshev polynomial roots are first obtained, and then the approximation coefficients C_k are calculated using the DCT of these points. Results have shown that the kind of Chebyshev polynomial and DCT that are used play an important role in the accuracy of the proposed method.

At the same time, it was shown in [20–22] that from an algebraic signal processing theory point of view, we can obtain different kinds of discrete cosine and sine transforms based on different signal models. For example, if we choose the space model for different kinds of Chebyshev polynomials, we obtain 16 kinds of discrete cosine and sine transforms [20]. The interpolation methods proposed in [18, 32] involve only one kind of DCT (DCT-2); therefore, research into interpolation that involves other kinds of DCTs/DSTs is interesting and worthwhile.

In this paper, we propose a new interpolation method for approximating the signal $\{g(t), t \in [0, L], g \in \{\mathcal{L}^2(0, L)\}\}$ based on the Chebyshev nonuniform sampling (CNS) points. The paper is organized as follows: preliminaries are reviewed in Sect. 2, the main results are presented in Sect. 3, the simulation results are described in Sect. 4, and the paper is concluded in Sect. 5.

2 Preliminaries

For a given real-valued function $\{g(t), t \in [0, L], g \in \{\mathcal{L}^2(0, L)\}\}$, the problem of Chebyshev interpolation based on DCT can be expressed as how to reconstruct this function from the Chebyshev nonuniform sampling points. This section reviews the existing results.

2.1 Previous Methods

There are two schemes of interpolation. One is given by Neagoe in [17, 18], and the other is given by Wang in [29–31]. In [18], suppose the j th root of the N th-order Chebyshev polynomial of the first kind is x_j , i.e.

$$x_j = -\cos \frac{(2j+1)\pi}{2N} = \cos \frac{(2(N-j)-1)\pi}{2N}, \quad (-1 < x_0 < x_1 < \dots < x_{N-1} < 1) \quad (1)$$

Then the nonuniform sampling grid vector $t_N = (t_j)_{j=0}^{N-1}$ is given based on x_j as follows

$$t_j = (L/2)(1 + x_j), \quad 0 \leq j \leq N-1. \quad (2)$$

The interpolation formula derived in [18] is

$$\tilde{g}_N(t) = \sum_{h=0}^{N-1} C_h \tilde{T}_h \left(2 \frac{t}{L} - 1 \right), \quad (3)$$

where C_h can be obtained from a matrix related to the roots of the N th-degree Chebyshev polynomial of the first kind, and $\tilde{T}_h(x)$ is the h th-degree normalized Chebyshev polynomial $\tilde{T}_0(x) = \frac{1}{\sqrt{2}}$,

$$\tilde{T}_h(x) = (T_h)(x) = \cos(h \arccos x), \quad h = 1, \dots, N-1. \quad (4)$$

It has been shown that the signal samples and the coefficients of the corresponding Chebyshev polynomial finite series are essentially a DCT pair [18].

In contrast to the method proposed in [18], Wang [29] proposed a method that replaces x_j in Eq. (1) with the roots of the N th-order Chebyshev polynomial of the second kind, i.e.

$$x_j = -\cos \frac{j\pi}{N} = \cos \frac{(N-j)\pi}{N}, \quad (-1 \leq x_0 < x_1 < \dots < x_N \leq 1) \quad (5)$$

The interpolation formula derived in [29] is then

$$\tilde{g}(t) = \sum_{k=0}^N C_k T_k \left(2 \frac{t}{L} - 1 \right), \quad (6)$$

where the Chebyshev coefficients C_k are given by

$$C_k = \frac{2(-1)^k}{N} \gamma_k \sum_{n=0}^{N-1} \gamma_n g(n) \cos \frac{kn\pi}{N}, \quad k = 0, 1, \dots, N, \quad (7)$$

where

$$\gamma_{i(i=k \text{ or } n)} = \begin{cases} \frac{1}{2}, & i = 0 \text{ or } i = N \\ 1, & \text{otherwise} \end{cases} \quad (8)$$

The main idea in [18] is to first derive the nonuniform sampling points and then apply the DCT to these samples to obtain the interpolation coefficient. However, [29] first proposed a modified definition of the first type of DCT and then calculated the values of C_k directly [29].

2.2 DCTs

A DCT expresses a finite sequence of data points in terms of a sum of cosine functions oscillating at different frequencies. DCTs are important for numerous applications in science and engineering, from lossy compression of audio and images to spectral methods for the numerical solution of partial differential equations [26]. DCTs are also closely related to Chebyshev polynomials, and fast DCT algorithms are used in the Chebyshev approximation of arbitrary functions by a series of Chebyshev polynomials, for example in Clenshaw–Curtis quadrature [25].

It has been shown that we can obtain 16 different kinds of discrete sine and cosine transforms [20], and the most common DCT variant is the type II DCT [25]. In fact, the interpolation methods proposed in [18, 29] involve only one kind of DCT (type II DCT). However, to the best of our knowledge, there is no published research that addresses interpolation based on the Chebyshev nonuniform sampling points associated with the other DCTs.

3 Main Results

Based on the existing results about Chebyshev polynomial interpolation based on DCTs, we propose a novel interpolation method that consists of two steps: (a) nonuniform points are sampled using the second kind of Chebyshev polynomial roots, and (b) a DCT is applied to the nonuniform samples to obtain the coefficients C_k .

Let x_j represent the j th root of the N th-order Chebyshev polynomial of the second kind, i.e.

$$x_j = -\cos \frac{j\pi}{N} = \cos \frac{(N-j)\pi}{N}. \quad (-1 \leq x_0 < x_1 < \cdots < x_N \leq 1) \quad (9)$$

The nonuniform sampling grid vector $\mathbf{t}_{N+1} = (t_j)_{j=0}^N$ is given by

$$t_j = (L/2)(1 + x_j), \quad 0 \leq j \leq N. \quad (10)$$

The main result of this paper is expressed in the following theorem.

Theorem 1 Let $\mathbf{g}_{N+1} = (g(t_j))_{j=0}^N$ be a vector of nonuniformly sampled signal points according to the Chebyshev sampling grid vector $\mathbf{t}_{N+1}$. The interpolation formula based on the Chebyshev nonuniform sampling points t_j can then be represented as

$$\tilde{g}(t) = \sum_{k=0}^N C_k T_k \left(2 \frac{t}{L} - 1 \right), \quad (11)$$

where the coefficients $C_k, k = 0, \dots, k = N$ are defined as

$$\mathbf{C}_{N+1} = \Psi_{N+1}^{DDCT} \times \mathbf{g}_{N+1}, \quad (12)$$

where $\mathbf{C}_{N+1} = (C_0, C_1, \dots, C_N)^T$ is the direct DCT (DDCT) of vector $\mathbf{g}_{N+1}$.

Proof

□

Consider the matrix

$$\Psi_{N+1}^{DCT} = (\psi_{mn})_{0 \leq m \leq N; 0 \leq n \leq N}, \quad (13)$$

and its elements, which characterize the DCT as

$$\psi_{mn} = (-1)^m \sqrt{\frac{2}{N}} k(m) k(n) \cos \frac{mn\pi}{N}, \quad (14)$$

where

$$k_{i(i=m \text{ or } n)} = \begin{cases} \frac{1}{\sqrt{2}}, & \text{if } i = 0 \text{ or } i = N \\ 1, & \text{if } i \neq 0 \text{ and } i \neq N \end{cases} \quad (15)$$

Therefore,

$$\mathbf{C}_{N+1} = \Psi_{N+1}^{DDCT} \times \mathbf{g}_{N+1}, \quad (16)$$

and

$$\Psi_{N+1}^{DDCT} = \sqrt{\frac{2}{N}} \Psi_{N+1}^{DCT}. \quad (17)$$

The inverse DCT (IDCT) of Eq. (16) can be expressed as

$$\mathbf{g}_{N+1} = \Psi_{N+1}^{IDCT} \times \mathbf{C}_{N+1}, \quad (18)$$

where

$$\Psi_{N+1}^{IDCT} = \left(\Psi_{N+1}^{DDCT} \right)^{-1} = \sqrt{\frac{N}{2}} \left(\Psi_{N+1}^{DCT} \right)^T. \quad (19)$$

The final interpolation formula can then be derived from Eqs. (16–19).

Using the results of Theorem 1, we can approximate the original signal in Eq. (11). This method is different from the existing methods in [18, 29] because it uses the roots of the second type of Chebyshev polynomial and connects the approximation using the first kind of DCT (type I DCT). Simulations and numerical results comparing the methods are presented in the following section.

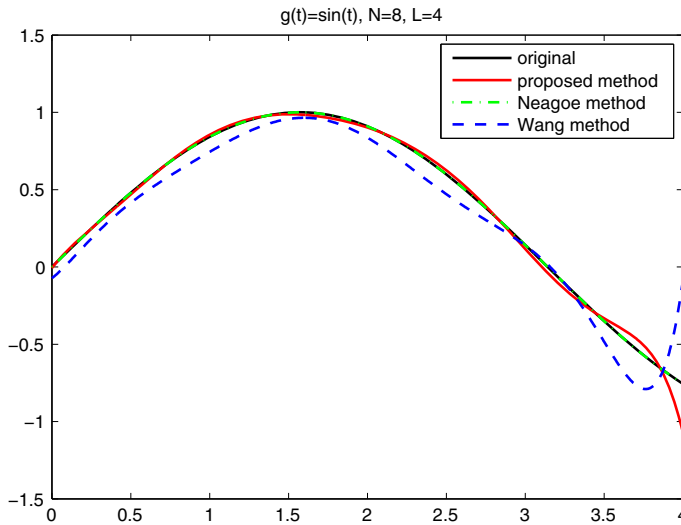


Fig. 1 Approximation results for $\sin(t)$ obtained by the Wang, Neagoe, and proposed methods

4 Simulations

To verify the method proposed in this paper, we used three test signals, $g(t) = \sin(t)$, $\cos(t)$, and $t^2 - 3t + 2$, for the simulation.

4.1 Simulation Results for Constant Signal Duration and Sampling Points

In this simulation, the signal duration L was $t \in [0, 4]$ and the number of sampling points $N = 8$. The interpolation results for the proposed method and classical methods (Wang [29] and Neagoe [18] methods) are plotted in Figs. 1, 2, 3, respectively.

From the simulation results, it can be concluded that the proposed method obtains good results when compared to the classical methods. To investigate in detail the errors between the original and reconstructed signals, an error analysis is presented in the following section.

4.2 Error Analysis

Two error analysis tools, maximum instantaneous error (MIE) and root mean squared (RMS) interpolation error, were used to evaluate the proposed and classical methods.

MIE is defined as

$$\varepsilon_N = \max_{t \in [0, L]} |g(t) - \tilde{g}_N(t)|, \quad (20)$$

and RMS is defined as

$$(\varepsilon_{\text{rms}})_c = \sqrt{\frac{1}{L} \int_0^L [g(t) - \hat{g}_N(t)]^2 dt}. \quad (21)$$

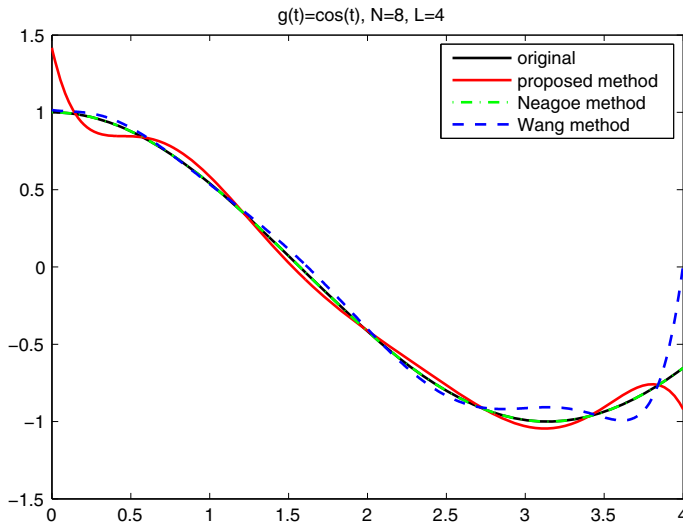


Fig. 2 Approximation results for $\cos(t)$ obtained by the Wang, Neagoe, and proposed methods

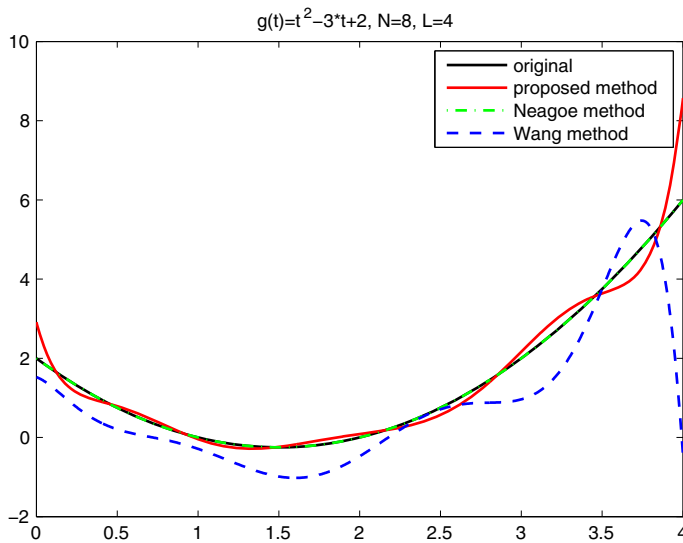


Fig. 3 Approximation results for $g(t) = t^2 - 3t + 2$ obtained by the Wang, Neagoe, and proposed methods

The results are compared in Tables 1, 2, 3 for the three test signals, respectively. We can see from the error analysis that the proposed method works better than the Wang method [29], but worse than the Neagoe method [18] for all test signals.

4.3 Influence of Sampling Rate

It was shown in Sects. 4.1 and 4.2 that the proposed method works well for different kinds of signals. In this section, we discuss the influence of sampling rate on the inter-

Table 1 Error analysis
[$\sin(t)$, $N = 8$, $L = 4$]

Error	Method in [18]	Method in [29]	Method in this paper
MIE	0.00004	0.68332	0.32158
RMS	0.00003	0.11155	0.03746

Table 2 Error analysis
[$\cos(t)$, $N = 8$, $L = 4$]

Error	Method in [18]	Method in [29]	Method in this paper
MIE	0.00003	0.66897	0.41792
RMS	0.00001	0.07942	0.06196

Table 3 Error analysis
[$t^2 - 3t + 2$, $N = 8$, $L = 4$]

Error	Method in [18]	Method in [29]	Method in this paper
MIE	0.00000	6.47595	2.57107
RMS	0.00000	0.87760	0.30284

Table 4 MIE error analysis
[$\sin(t)$, $N = 8$]

L	Method in [18]	Method in [29]	Method in this paper
2	0.00000	1.08124	0.38639
4	0.00004	0.68332	0.32158
8	0.00791	1.05928	0.41789
16	0.72524	0.32504	0.36358

polation error of signal $\sin(t)$ for various signal durations L and number of sampling points N .

In this analysis, we fix either the value of L or N and vary the other parameter. The simulation results are given in Tables 4, 5, 6, 7. From these results, we conclude that the performance of the Wang method improves as signal duration L increases, while the performance of the Neagoe method degrades. The performance of the proposed method lies between the other two methods. All these methods work well as the number of sampling points increases.

Table 5 RMS error analysis
[$\sin(t)$, $N = 8$]

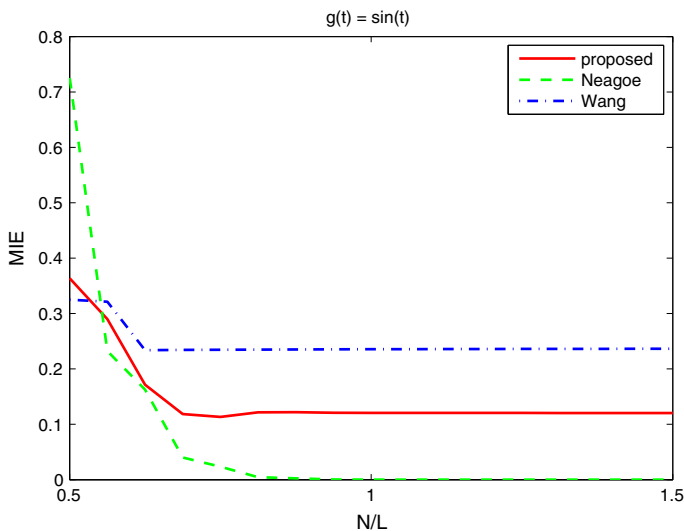
L	Method in [18]	Method in [29]	Method in this paper
2	0.00000	0.20855	0.04500
4	0.00003	0.11155	0.03746
8	0.00457	0.14023	0.04817
16	0.38560	0.13758	0.19270

Table 6 MIE error analysis
[$\sin(t)$, $L = 4$]

N	Method in [18]	Method in [29]	Method in this paper
4	0.07023	0.66946	0.30410
8	0.00004	0.68332	0.32158
16	0.00000	0.69025	0.31754
32	0.00000	0.69371	0.31551

Table 7 RMS error analysis
[$\sin(t)$, $L = 4$]

N	Method in [18]	Method in [29]	Method in this paper
4	0.04475	0.16494	0.07069
8	0.00003	0.11155	0.03746
16	0.00000	0.08134	0.02047
32	0.00000	0.06842	0.01148

**Fig. 4** MIE errors for varying N/L

Finally, we investigate the influence of the sampling rate N/L on the interpolation error of test signal $\sin(t)$. Figures 4 and 5 plot the trends of the MIE and RMS errors for varying sampling rates. This simulation also indicates that the interpolation errors decrease as the sampling rates increase, and the proposed method can be viewed as a compromise between the Wang and Neagoe methods.

5 Conclusion

A new approximation and interpolation method based on Chebyshev nonuniform sampling points and DCT was proposed in this paper. The proposed method applies the

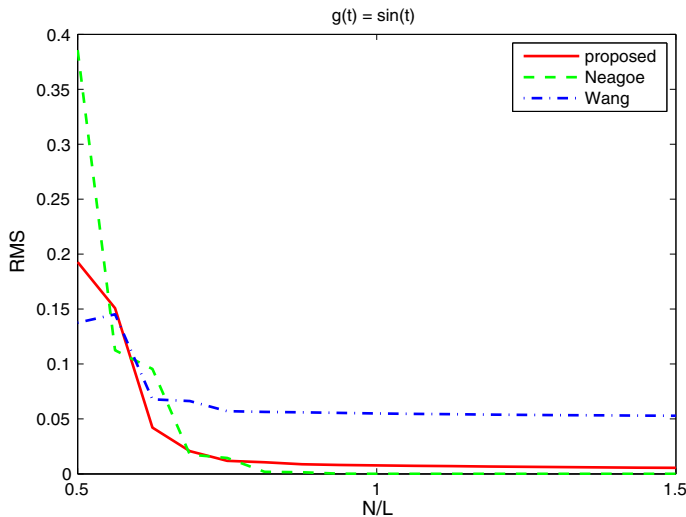


Fig. 5 RMS errors for varying N/L

nonuniform roots of the second type of Chebyshev polynomial and uses the type I DCT to obtain the interpolation coefficients. This method can be seen as a supplemental method to the classical results. Simulations and a numerical analysis were also carried out to verify the derived results. The derivation of an analytical formula for the interpolation error and an extension of the proposed method to more kinds of signals such as images and multidimensional signals are both directions for our future work.

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References

1. B. Adcock, A.C. Hansen, A. Shadrin, A stability barrier for reconstructions from Fourier samples. *SIAM J. Numer. Anal.* **52**(1), 125–139 (2014)
2. J.I. Agbinya, Interpolation using the discrete cosine transform. *Electron. Lett.* **28**(20), 1928 (1992)
3. M. Berzins, Adaptive polynomial interpolation on evenly spaced meshes. *SIAM Rev.* **49**(4), 604–627 (2007)
4. R.L. Burden, J.D. Faires, *Numerical Analysis* (Brooks Cole, Boston, 2010)
5. E.W. Cheney, *Approximation Theory III* (Academic Press Inc., New York, 1980)
6. A. Dutt, M. Gu, V. Rokhlin, Fast algorithms for polynomial interpolation, integration, and differentiation. *SIAM J. Numer. Anal.* **33**(5), 1689–1711 (1996)
7. L. Fax, I.B. Parker, *Chebyshev Polynomials in Numerical Analysis* (Oxford University Press, London, 1968)
8. K. Glashoff, K. Roleff, A new method for Chebyshev approximation of complex-valued functions. *Math. Comput.* **36**(153), 233–239 (1981)
9. M.A. Hernandez, Chebyshev's approximation algorithms and applications. *Comput. Math. Appl.* **41**(3), 433–445 (2001)
10. M.F. Huber, Chebyshev polynomial Kalman filter. *Digit. Signal. Process.* **23**(5), 1620–1629 (2013)

11. C.P. Li, B.Z. Li, T.Z. Xu, Approximating bandlimited signals associated with the LCT domain from nonuniform samples at unknown locations. *Signal Process.* **92**(7), 1658–1664 (2012)
12. B.Z. Li, T.Z. Xu, Spectral analysis of sampled signals in the linear canonical transform domain. *Math. Probl. Eng.* **2012**, 1–19 (2012)
13. P.S. Malachivskyy, Y.V. Pizyur, N.V. Danchak, E.B. Orazov, Chebyshev approximation by exponential-power expression. *Cybern. Syst. Anal.* **49**(6), 877–881 (2013)
14. Y. Mao, Z. Yu, Interpolation using the type I discrete cosine transform with greatly increased accuracy. *Electron. Lett.* **31**(2), 85–86 (1995)
15. I.I. Marks, J. Robert, *Introduction to Shannon Sampling and Interpolation Theory* (Springer, New York, 1991)
16. P.P. Massopust, *Interpolation and Approximation with Splines and Fractals* (Oxford University Press, London, 2010)
17. V.E. Neagoe, A two-dimensional nonuniform sampling expansion model. *Signal Process.* **33**(1), 1–21 (1993)
18. V.E. Neagoe, Chebyshev nonuniform sampling cascaded with the discrete cosine transform for optimum interpolation. *IEEE Trans. Acoust. Speech Signal Process.* **38**(10), 1812–1815 (1990)
19. D. Potts, M. Tasche, Sparse polynomial interpolation in Chebyshev bases. *Linear Algebra Appl.* **441**, 61–87 (2014)
20. M. Puschel, J.F. Moura, Algebraic signal processing theory. ArXiv preprint. [cs/0612077](https://arxiv.org/abs/cs/0612077)(2006)
21. M. Puschel, J.F. Moura, Algebraic signal processing theory: foundation and 1-D time. *IEEE Trans. Signal Process.* **56**(8), 3572–3585 (2008)
22. M. Puschel, J.F. Moura, Algebraic signal processing theory: 1-D space. *IEEE Trans. Signal Process.* **56**(8), 3586–3599 (2008)
23. T.J. Rivlin, *Chebyshev Polynomials: From Approximation Theory to Algebra and Number Theory* (Wiley, New York, 1990)
24. L.A. Sakhnovich, *Interpolation Theory and Its Applications* (Springer, New York, 1997)
25. G. Strang, The discrete cosine transform. *SIAM Rev.* **41**(1), 135–147 (1999)
26. L.N. Trefethen, Is Gauss quadrature better than Clenshaw–Curtis? *SIAM Rev.* **50**(1), 67–87 (2008)
27. L.N. Trefethen, *Approximation Theory and Approximation Practice* (Society for Industrial and Applied Mathematics, Philadelphia, 2013)
28. H. Triebel, *Interpolation Theory, Tunction Spaces, Differential Operators* (Wiley-VCH, Johann Ambrosius Barth, Heidelberg, 1995)
29. Z. Wang, Interpolation using type I discrete cosine transform. *Electron. lett.* **26**(15), 1170–1172 (1990)
30. Z. Wang, Interpolation using the discrete cosine transform: reconsideration. *Electron. Lett.* **29**(2), 198–200 (1993)
31. Z. Wang, G.A. Jullien, W.C. Miller, Fast computation of Chebyshev optimal nonuniform interpolation. *Proceedings of the 38th Midwest Symposium on Circuits and Systems.* **1**, 111–114 (1995)
32. Z. Wang, L. Wang, Interpolation using the fast discrete sine transform. *Signal Process.* **26**(1), 131–137 (1992)
33. T.Z. Xu, B.Z. Li, *Linear Canonical Transform and its Applications* (Science Press, Beijing, 2013)
34. A. Zakhor, G. Alvstad, Two-dimensional polynomial interpolation from nonuniform samples. *IEEE Trans. Signal Process.* **40**(1), 169–180 (1992)